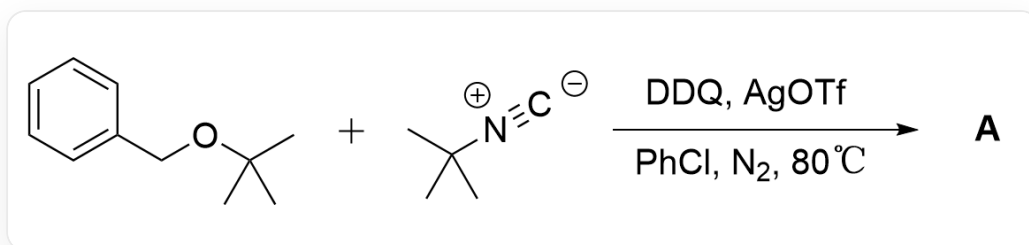


## 题目

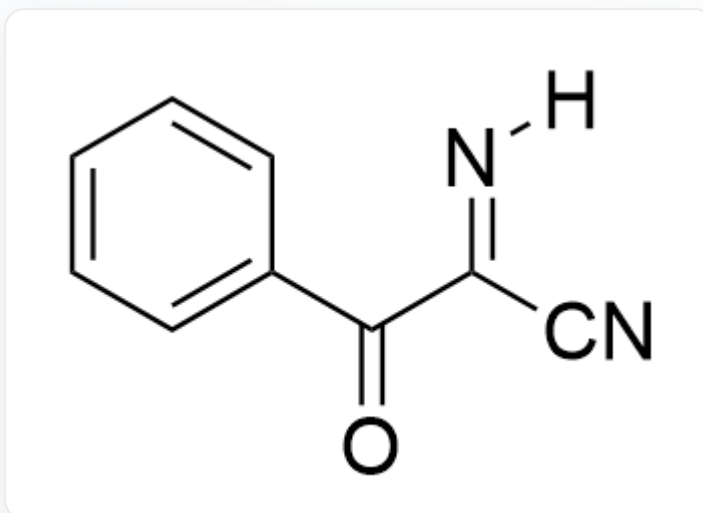


CC(C)(C)OCC1=CC=CC=C1.CC(C)([N+]\#C-)>[DDQ],[AgOTf],[PhCl],[N2]>[A],A为反应产物,反应温度为80°C

其中DDQ和叔丁基异氰的量至少是苄基叔丁基醚的两倍，机理研究表明该反应并非自由基反应机理，试推断出反应产物 **A** 的结构式（不考虑立体异构）

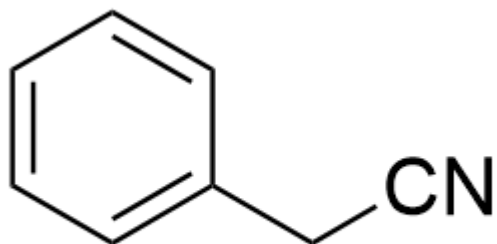
**A.** 其他选项均不正确

**B.**



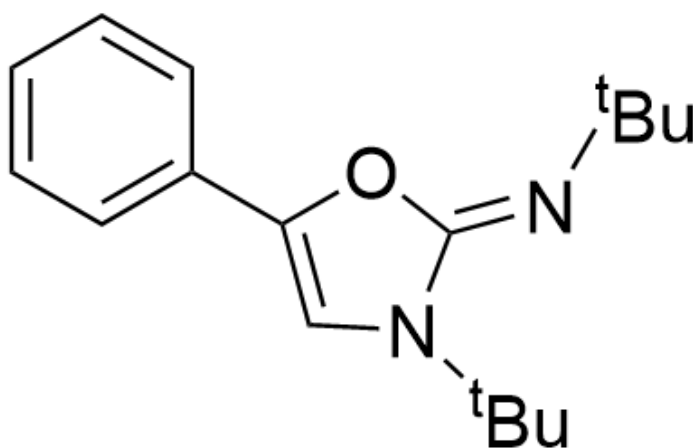
[H]/N=C(C#N)/C(C1=CC=CC=C1)=O

**C.**



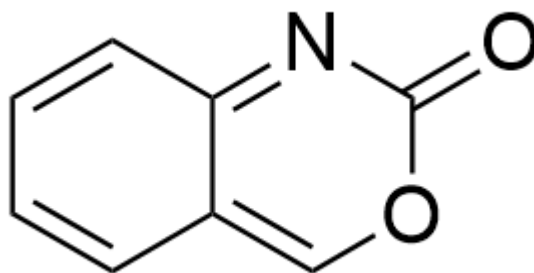
N#CCCC1=CC=CC=C1

D.



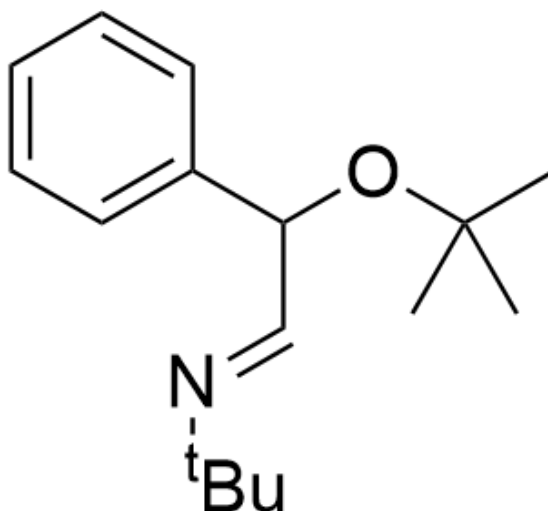
CC(/N=C1N(C(C)(C)C)C=C(O1)C2=CC=CC=C2)(C)C

E.



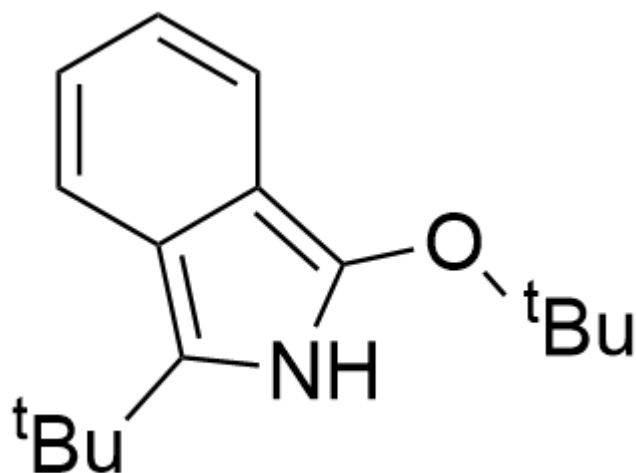
O=C1OC=C2C=CC=CC2=N1

F.



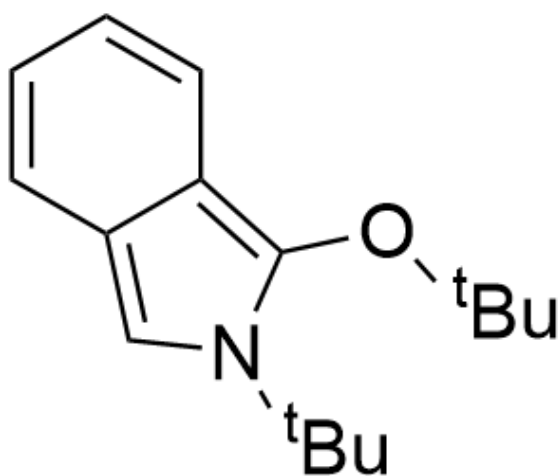
CC(C)(C)OC(/C=N/C(C)(C)C)C1=CC=CC=C1

G.



CC(OC1=C2C(C=CC=C2)=C(C(C)(C)C)N1)(C)C

H.



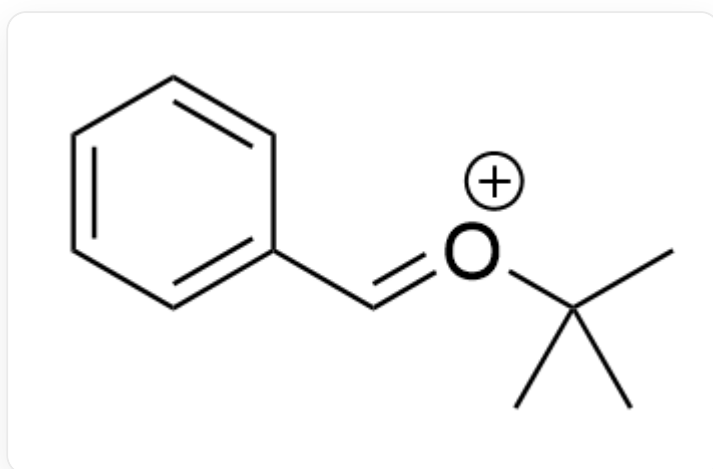
CC(OC1=C2C(C=CC=C2)=CN1C(C)(C)C)(C)C

## 答案

正确答案: **A**

## 详细解析

首先苄醚底物被DDQ氧化形成中间体1



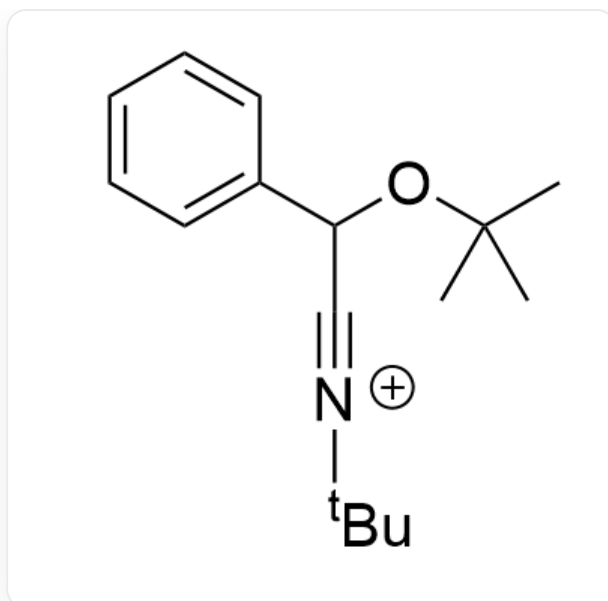
CC(C)(C)/[O+]=C/C1=CC=CC=C1

### CHECKPOINT

1 PTS

CC(C)(C)/[O+]=C/C1=CC=CC=C1

随后发生一步加成反应得到中间体2



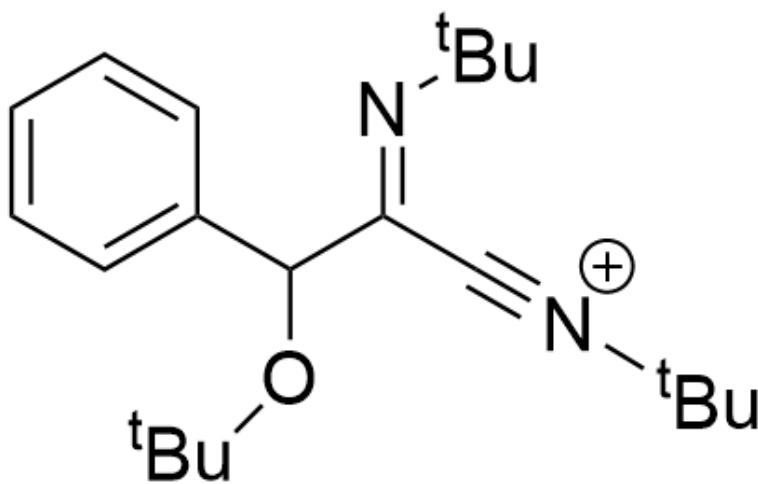
CC(C)(C)OC(C#[N+](C(C)(C)C)C1=CC=CC=C1

CHECKPOINT

1 PTS

CC(C)(C)OC(C#[N+](C(C)(C)C)C1=CC=CC=C1

再次发生一分子加成反应得到中间体3



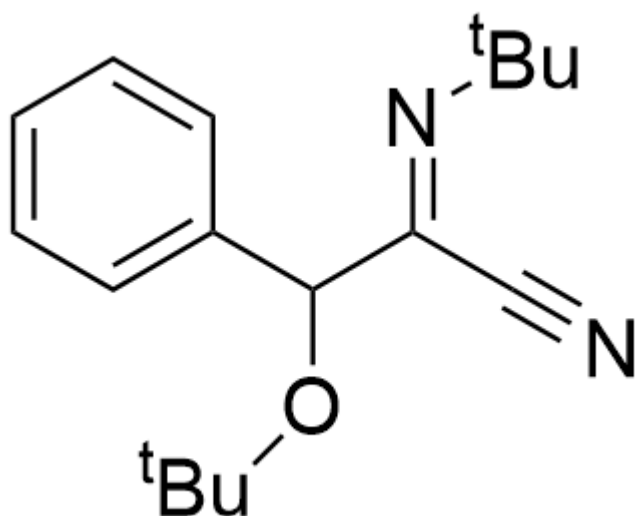
CC(OC(/C(C#[N+]C(C)(C)C)=N/C(C)(C)C)C1=CC=CC=C1)(C)C

CHECKPOINT

1 PTS

CC(OC(/C(C#[N+]C(C)(C)C)=N/C(C)(C)C)C1=CC=CC=C1)(C)C

随后脱去叔丁基正离子得到中间体4



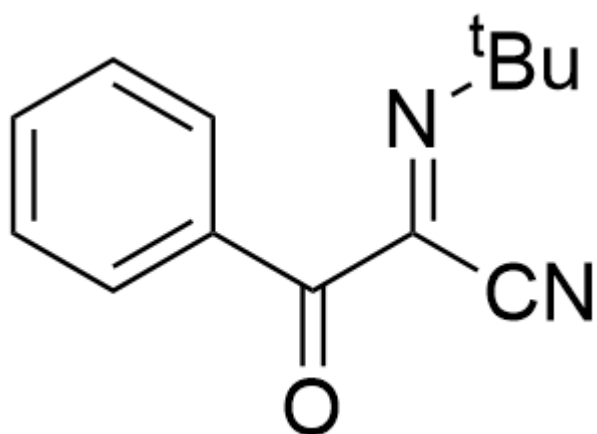
N#C/C(C(OC(C)(C)C)C1=CC=CC=C1)=N\C(C)(C)C

CHECKPOINT

1 PTS

N#C/C(C(OC(C)(C)C)C1=CC=CC=C1)=N\C(C)(C)C

最后再被一分子 DDQ 氧化得到反应产物 **A**



O=C(/C(C#N)=N/C(C)(C)C)C1=CC=CC=C1

## CHECKPOINT

1 PTS

O=C(/C(C#N)=N/C(C)(C)C)C1=CC=CC=C1